

Math 271: Mathematical Methods (Fall 2026)

Model Solution for Class Test 2, held on 18 August 2026

Throughout, $A = (a_{ij})$ is an $m \times n$ matrix, $A^1, \dots, A^n \in \Re^m$ are its columns, $\text{rank}(A) = k$, and A_{ij} denotes the cofactor of the (i, j) entry of whatever square matrix is under discussion. Furthermore, the following results may be quoted from the class notes without proof, and no points are lost for doing so, however, for an application of a theorem or lemma, verification of applicability conditions are met to receive full credit:

- (R1) the Rank Theorem (row rank = column rank);
 - (R2) for a square matrix M : $|M| \neq 0 \iff M \text{ non-singular} \iff \text{rank}(M) \text{ is full} \iff$ the columns of M are linearly independent;
 - (R3) the expansion of a determinant along any row or any column, $|M| = \sum_j m_{ij} M_{ij}$;
 - (R4) the vanishing of expansions by alien cofactors, $\sum_j m_{ij} M_{kj} = 0$ for $i \neq k$;
 - (R5) the elementary properties of determinants: $|M| = |M^\top|$, behaviour under interchange of two rows (or columns), a determinant with two equal or proportional rows (or columns) is zero, invariance under adding a multiple of one row to another.
 - (R6) Fundamental Theorem on Vector Spaces
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Since $\text{rank}(A) = k$ we can choose k linearly independent columns of A . Both the antecedent and the consequent to be proven are unchanged if the columns of A are renumbered and the rows of A are permuted:

1. Independence of a set of columns does not depend on the order in which its members are listed, and,
2. an interchange of two rows or two columns multiplies each minor of a given order by ± 1 while permuting the minors of that order among themselves. Since the statement to be proved quantifies over all minors of a given order, it is therefore invariant under such relabellings, and we may assume that the k independent columns selected are the first k .

Thus, we define $\mathcal{A}^C = \{A^1, \dots, A^k\}$ as the set of the k linearly independent columns. Clearly, any column of A , say A^j is a linear combination of $\mathcal{A}^C = \{A^1, \dots, A^k\}$, for otherwise, the $k+1$ columns, $\mathcal{A}^C \cup \{A^j\}$ would be independent, contradicting $\text{rank}(A) = k$. Hence, we can write

$$A^j = \sum_{s=1}^k \alpha_{sj} A^s \quad \forall j = 1, \dots, n \quad (1)$$

Let M be any sub-matrix of order $k+1$, formed by rows $r_1 < \dots < r_{k+1}$ and columns $q_1 < \dots < q_{k+1}$, and for $v \in \Re^m$ let $\pi(v)$ denote $(v_{r_1}, \dots, v_{r_{k+1}})^\top \in \Re^{k+1}$. The map $\pi : \Re^m \rightarrow \Re^{k+1}$ is defined component-wise, and hence, preserves linear combinations.

$$\pi(\lambda v + \mu w) = \lambda \pi(v) + \mu \pi(w)$$

Applying π to (1) therefore leaves the coefficients unchanged, and reading the identity (1) in the coordinates $r_1, \dots, r_{k+1}$ only, we get

$$M^j = \pi(A^{qj}) = \sum_{s=1}^k \alpha_{sqj} \pi(A^s) \text{ for } j = 1, \dots, k+1$$

The $k+1$ columns of M are linear combinations of k vectors of $\Re^{k+1}$. Hence, from the *Fundamental Theorem on Vector Spaces*¹

If each of the $(q+1)$ vectors $y^0, y^1, \dots, y^q$ in $\Re^d$ can be expressed as a linear combination of the p vectors $x^1, x^2, \dots, x^p$ in $\Re^d$ then the vectors $y^0, y^1, \dots, y^q$ are linearly dependent.

Setting $k = q$, each column M^j as the y vectors and $\pi(A^j)$ as the x vectors, we get that columns of M are linearly dependent. Hence, M is singular, whence, $|M| = 0$ by (R2). M was an arbitrary submatrix of order $k+1$ - its rows and columns are both arbitrary, hence *every* minor of order $k+1$ of A vanishes.

Now we will show that there is at least one minor of order k that does not vanish.

Assume, to the contrary, that all determinants of order k vanish. We call this the contrary hypothesis and refer to it in shorthand (H)

Let C be the $m \times k$ matrix formed by $\mathcal{A}^C$, that is $C = (A^1, A^2, \dots, A^k)$ and suppose every determinant of order k formed from them vanishes.

Let $M(\mathcal{A}^C)$ be the top $k \times k$ block of C and $A_{k1}, \dots, A_{kk}$ the cofactors of its last row.

Expanding $|M(\mathcal{A}^C)|$ along its last row,

$$\sum_{i=1}^k a_{ki} A_{ki} = |M(\mathcal{A}^C)| = 0$$

where the first equality follows from the expansion of the determinant along its k^{th} row (R3) and the second equality follows from the contrary hypothesis, as $M(\mathcal{A}^C)$ is a $k \times k$ submatrix formed from the first k columns of A

For $\ell > k$, let $M(A^\ell)$ be the $k \times k$ submatrix of C with rows $1, \dots, k-1, \ell$; then $|M(A^\ell)| = 0$ by contrary hypothesis, and expanding along its last row, whose entries are $a_{\ell 1}, \dots, a_{\ell k}$, gives

$$\sum_{i=1}^k a_{\ell i} A_{ki} = 0$$

This is because:

1. SAME MINOR: deleting the last row and the i -th column of $M(A^\ell)$ leaves exactly the same matrix as deleting the last row and the i -th column of $M(\mathcal{A}^C)$: a cofactor of an

¹Refer Homework 1, Question 3 and/or Theorem 1 from Lecture Notes on Vectors

entry in the last row never looks at the last row, and the two matrices, $M(\mathcal{A}^C)$ and $M(A^\ell)$ differ only in their last row:

$$M(\mathcal{A}^C) = \begin{pmatrix} a_{11} & \cdots & a_{1k} \\ \vdots & & \vdots \\ a_{k-1,1} & \cdots & a_{k-1,k} \\ a_{k1} & \cdots & a_{kk} \end{pmatrix}, \quad M(A^\ell) = \begin{pmatrix} a_{11} & \cdots & a_{1k} \\ \vdots & & \vdots \\ a_{k-1,1} & \cdots & a_{k-1,k} \\ a_{\ell 1} & \cdots & a_{\ell k} \end{pmatrix} \quad (\ell > k).$$

2. SAME SIGN given by $(-1)^{k+i}$:

We compare the two matrices. Write N_i for the $(k-1) \times (k-1)$ matrix consisting of rows $1, \dots, k-1$ of C with column i removed.

- In $M(\mathcal{A}^C)$ the entry a_{ki} occupies position (k, i) : deleting row k and column i leaves N_i , so its cofactor is $A_{ki} = (-1)^{k+i}|N_i|$.
- In $M(A^\ell)$ the entry $a_{\ell i}$ also occupies position (k, i) – it is in the k -th (last) row of that $k \times k$ matrix, even though the numbers in it were taken from row ℓ of C . Deleting row k and column i again leaves N_i (the first $k-1$ rows are the same in both matrices), so its cofactor is $(-1)^{k+i}|N_i|$.

For $\ell < k$, the sum $\sum_{i=1}^k a_{\ell i} A_{ki}$ pairs the entries of row ℓ of $M(\mathcal{A}^C)$ with the cofactors of its row k . Such an expansion by alien cofactors vanishes for every matrix, by (R4) — no appeal to (H) is needed here — so $\sum_{i=1}^k a_{\ell i} A_{ki} = 0$

The three cases $\ell = k, \ell > k$ and $\ell < k$ exhaust $\ell = 1, \dots, m$, and together they say $\sum_{i=1}^k A_{ki} A^i = \mathbf{0}$; because the j -th coordinate of the left-hand side is $\sum_{i=1}^k A_{ki} a_{ji}$ — if the A_{ki} were not all zero this would be a non-trivial linear combination of $A^1, \dots, A^k$, contradicting their independence.

Hence all A_{ki} vanish; each is $\pm$ a determinant of order $k-1$ formed from the first k columns, and by rearranging the rows and the first k columns any non-vanishing determinant of order $k-1$ could be brought into that position. So all determinants of order $k-1$ formed from those columns vanish as well, and the argument may be repeated with $k-1$ in place of k .

The descent ends with two of the chosen columns u, v all of whose 2×2 minors vanish: $u_i v_j - u_j v_i = 0$ for all i, j . If $u = \mathbf{0}$, then u, v are linearly dependent and the determinant vanishes trivially. Otherwise choose p with $u_p \neq 0$ and set the value $\mu = v_p/u_p$; taking $j = p$ in the hypothesis gives $v_i = \mu u_i$ for every i , i.e. $v = \mu u$, so $\mu u - v = \mathbf{0}$ is a non-trivial relation and u, v are dependent. But two vectors from a set of linearly independent vectors $\mathcal{A}^C$ are independent — a contradiction.

Therefore not all determinants of order k formed from k independent columns can vanish: at least one minor of order k of A is non-zero. $\square$